

DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS
PH1020 Physics II
Lecture Note-1

Solutions of Laplace's Equation Date: 26 Jan 2025

1. Laplace's Equation

Given a stationary charge distribution $\rho(\boldsymbol{r})$ we can, in principle, calculate the the electrostatic potential using the formula

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{1}{r}\rho\left(\mathbf{r}'\right) d\tau'.$$

The electrostatic potential $V(\mathbf{r})$ can only be evaluated analytically for the simplest charge configurations using above formula. Also in many electrostatic problems, conductors are involved and the charge distribution $\rho(\boldsymbol{r})$ is not known in advance (only the total charge on each conductor is known).

A better approach to determine the electrostatic potential is to start with Poisson's equation

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}$$

Very often we only want to determine the potential in a region where $\rho = 0$. In this region Poisson's equation reduces to Laplace's equation

$$\nabla^2 V = 0$$

There are an infinite number of functions that satisfy Laplace's equation and the appropriate solution is selected by specifying the appropriate boundary conditions.

2. Solution of Laplace's Equation in Cartesian Coordinates (1D case)

In one dimension the electrostatic potential V depends on only one variable x. The electrostatic potential $V(x)$ is a solution of the one-dimensional Laplace equation

$$\frac{d^2V}{dx^2} = 0$$

The general solution of this equation is a straight line

$$\boxed{V(x) = mx + b}$$

where m and b are arbitrary constants. These constants are fixed when the value of the potential is specified at two different positions.

The following properties are true for any solution of the one-dimensional Laplace equation:

Property-1: $V(x)$ is average of $V(x+a)$ and $V(x-a)$ for any a

$$V(x) = \frac{1}{2}[V(x+a) + V(x-a)]$$

Laplace equation is kind of averaging instruction.

Property-2: The solution of Laplace's equation can not have local maxima or minima. Extreme values must occur at the end points (the boundaries). This is a direct consequence of property-1.

Property-2 has an important consequence: A charged particle can not be held in stable equilibrium by electrostatic forces alone (**Earnshaw's Theorem**). A particle is in a stable equilibrium if it is located at a position where the potential has a minimum value.

A small displacement away from the equilibrium position will increase the electrostatic potential of the particle, and a restoring force will try to move the particle back to its equilibrium position. However, since there can be no local maxima or minima in the electrostatic potential, the particle can not be held in stable equilibrium by just electrostatic forces.

3. Solution of Laplace's Equation in Spherical Coordinates (1D case)

Laplace's equation is

$$\nabla^2 V = 0$$

Laplace's equation in spherical coordinates is given by

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 V}{\partial\varphi^2} = 0$$

If V is only a function of r then

$$V \equiv V(r)$$
$$\frac{\partial V}{\partial\theta} = 0$$
$$\frac{\partial V}{\partial\varphi} = 0$$

Therefore, Laplace's equation can be rewritten as

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) = 0$$
$$r^2\frac{\partial V}{\partial r} = a = \text{const}$$

This differential equation can be rewritten as

$$\frac{\partial V}{\partial r} = \frac{a}{r^2}$$

The general solution of this first-order differential equation is

$$\boxed{V(r) = -\frac{a}{r} + b}$$

where b is a constant. If $V = 0$ at infinity ($V(\infty) = 0$) then b must be equal to zero , and consequently

$$V(r) = -\frac{a}{r}$$

4. Solution of Laplace's Equation in cylindrical Coordinates (1D case)

Laplace's equation is

$$\nabla^2 V = 0$$

Laplace's equation in Cylindrical coordinates is

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \varphi^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

If V is only a function of r then

$$V \equiv V(r)$$
$$\frac{\partial V}{\partial \varphi} = 0$$
$$\frac{\partial V}{\partial z} = 0$$

Therefore, Laplace's equation can be rewritten as

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0$$
$$r\frac{\partial V}{\partial r} = a = \text{const}$$

This differential equation can be rewritten as

$$\frac{\partial V}{\partial r} = \frac{a}{r}$$

The general solution of this first-order differential equation is

$$\boxed{V(r) = a\ln(r) + b}$$

where b is a constant. The constants a and b are determined by the boundary conditions.
[1]

[1]Copyright @ 2025 Department of Physics IIT Madras, Course Coordinator: Prasanta Kumar Muduli and AR Ganesan